Network Security Monitoring with NTP, Syslog, NetFlow and SNMP

ICT2203 Networks Security

LAB 9 NOTES

2021-2022 Trimester 1

LEARNING OUTCOMES

In this lab you will learn NTP and logging for network security monitoring

9.1 *Part1, Part2:*

NTP is for synchronizing time between network devices to facilitate analysis of log data, and can be implemented with security options to prevent attacks:

- MD5 authentication
- ACL

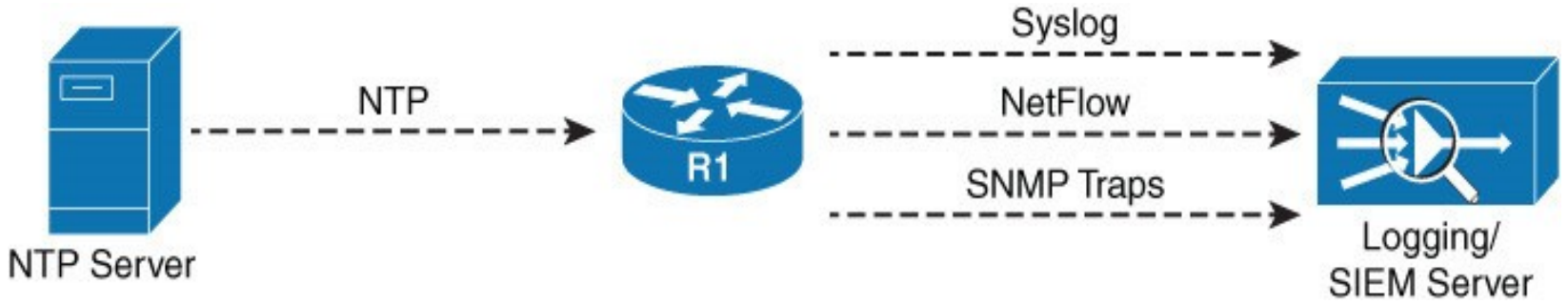
9.1 *Part 3:* Syslog is one of the most common method for logging

9.2 SNMP is not only used for logging, but also for managing network devices remotely

9.3 NetFlow is designed to provide statistical records of different IP packets that flow through a router

Finally, with the various network security implemented, we are now ready to perform network security monitoring by logging all alerts to centralized servers for analysis and reporting.

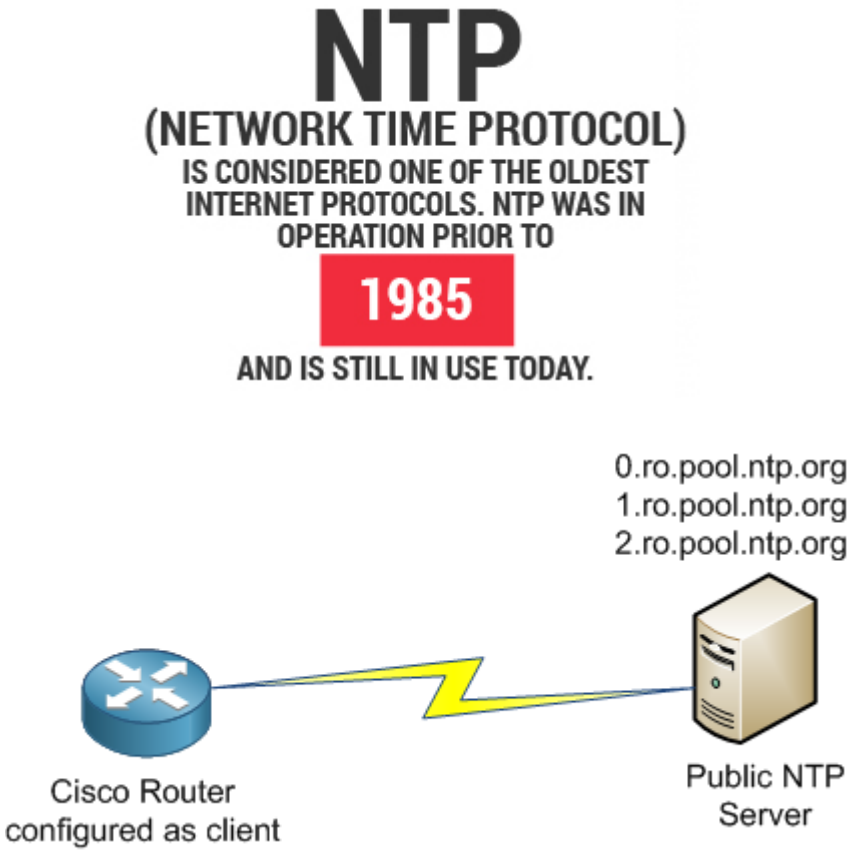
Logging is commonly implemented using Syslog, SNMP and/or NetFlow protocols to external servers or SIEM (Security Information and Event Management) systems.



To facilitate analysis of log data, it is essential to synchronize time between all network devices, which can be achieved by using NTP (Network Time Protocol).

(9.1 – *Part 1, 2*) NTP Overview

- **NTP** (**Network Time Protocol**) is used to allow network devices to synchronize their clocks with a central source clock.
  - If you ever have network issues or get hacked, you want to make sure you know exactly what and when it happened
- **What happens without NTP??**
  - Without time synchronization, each device will think the correct time is different
  - None of the timestamps line up when comparing logs
  - Not able to process the data for unusual behavior or indicators of compromise
  - Attackers may even tamper with NTP, knowing that it will complicate forensic log analysis



NTP OPERATION MODES

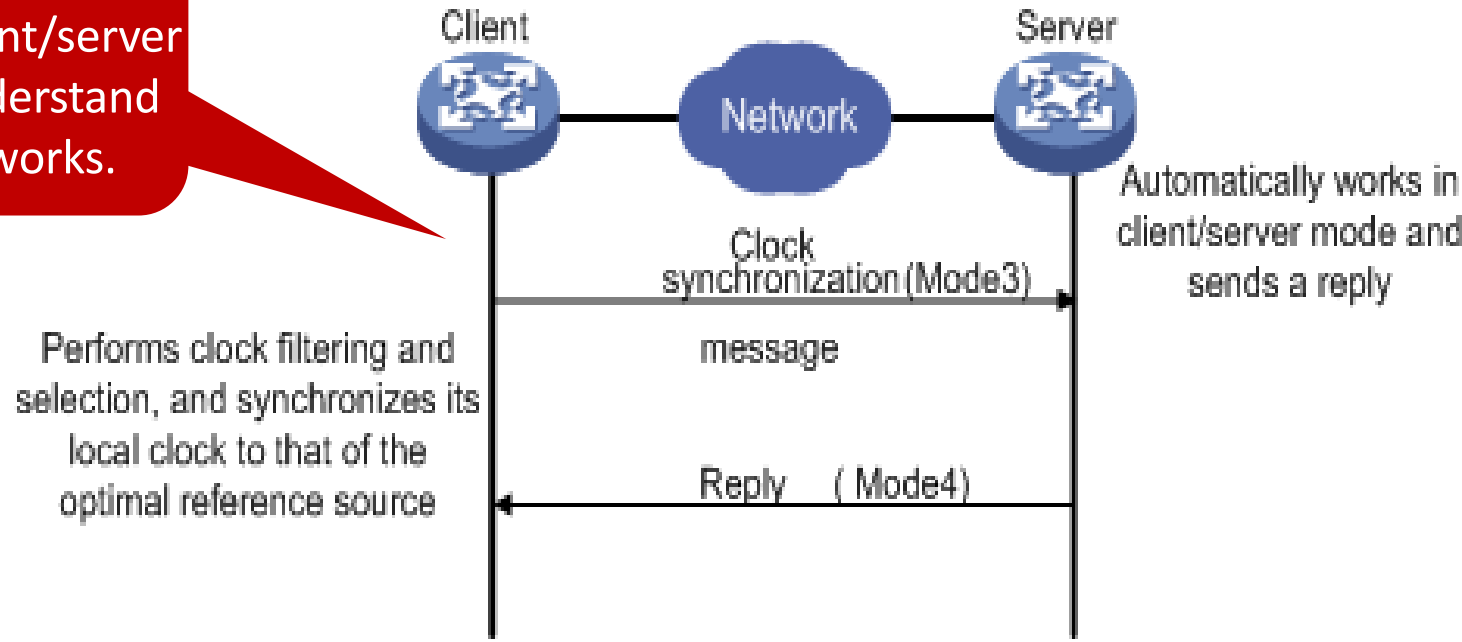
As defined in RFC 5905, there are 3 NTP operation modes

- *Client/server*,
- *Peer* &
- *Broadcast*

We'll discuss the common client/server mode to understand how NTP works.

NTP operates over UDP port 123

In client/server mode, the NTP client initiates a request to the NTP server, and receives a reply as shown:



In client/server mode, NTP client synchronizes its clock by performing the following computation based on NTP reply

Offset of client clock from server clock: $O = \frac{(T2 - T1) + (T3 - T4)}{2}$

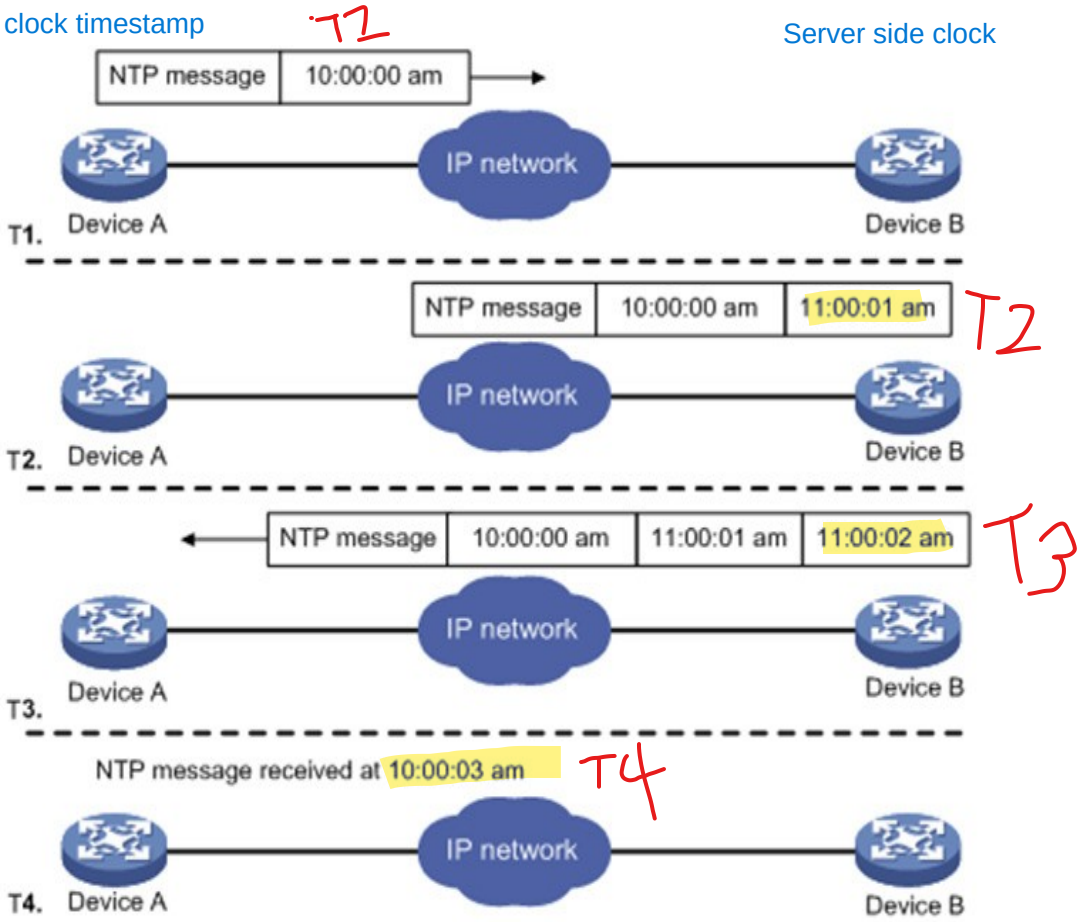
e.g. $O = \frac{1h1s + 59m59s}{2} = 1h$

Round-trip delay: $o = (T4 - T1) - (T3 - T2)$

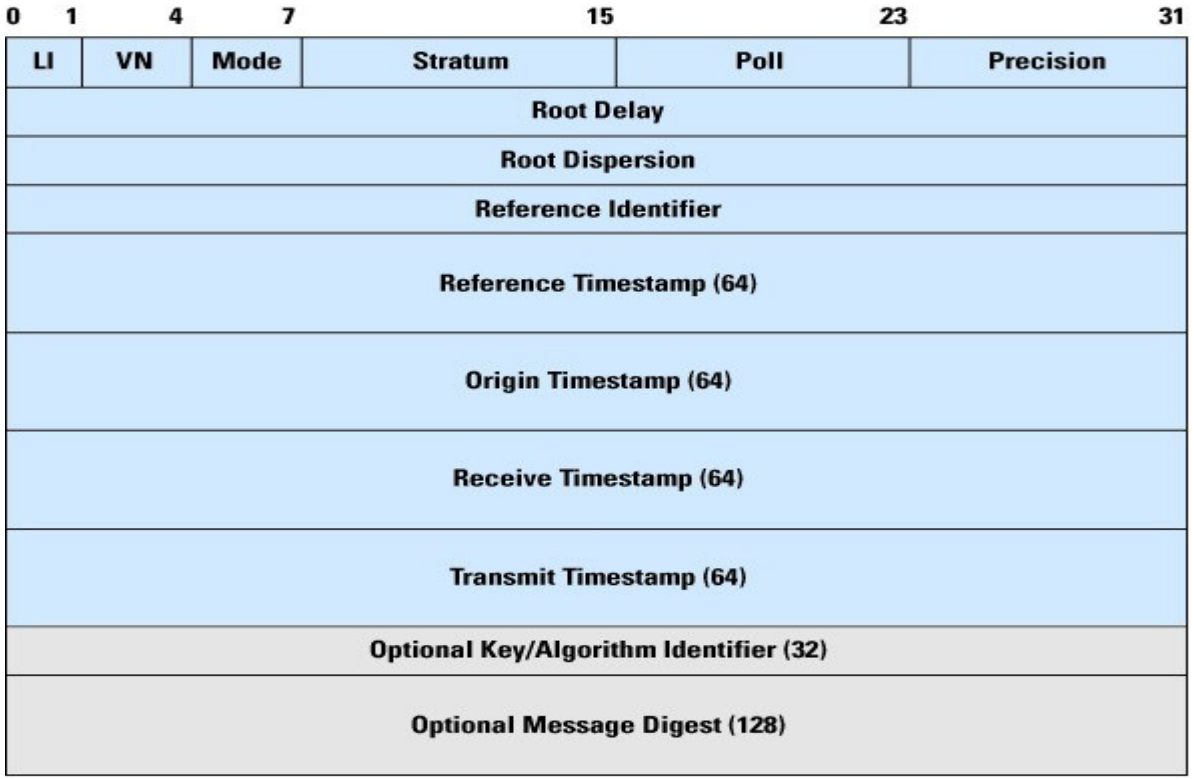
e.g. $O = 3s - 1s = 2s$

Round trip delay - is the delay time from client to the server
- smallest round trip delay is the server closest to client

(Clock filter and mitigation algorithms will then be used to determine the best and final offset to synchronize the system clock.)



To support NTP computation, NTP client and server register its timestamps in the NTP message format as follows:



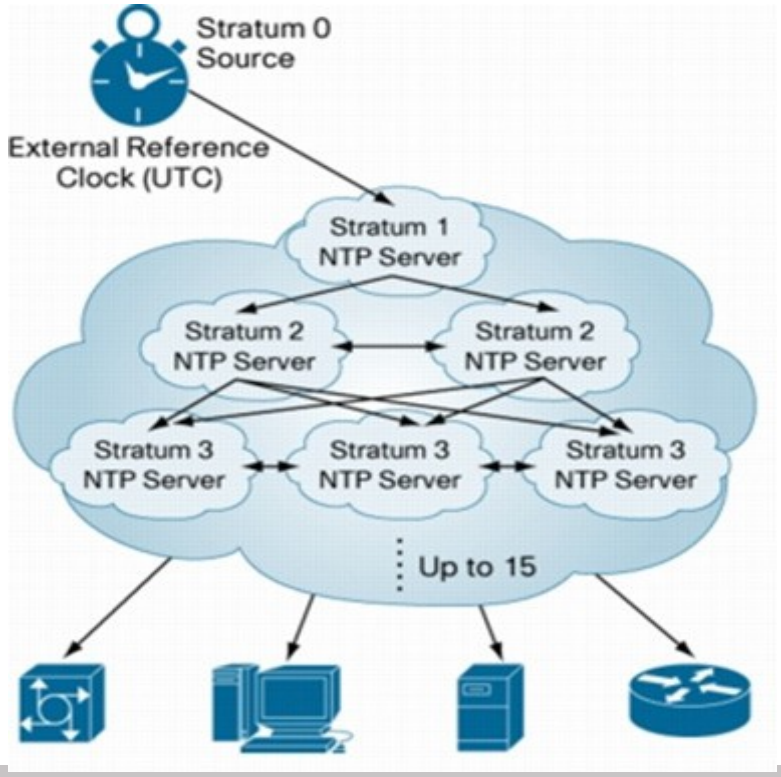
	NTP request	NTP reply
Origin timestamp:	ignore	T1
Receive timestamp:	ignore	T2
Transmit timestamp:	T1	T3

Reference timestamp: time when system clock was last updated

To be scalable, NTP servers are typically setup in a hierarchical structure to support time synchronization in practice.

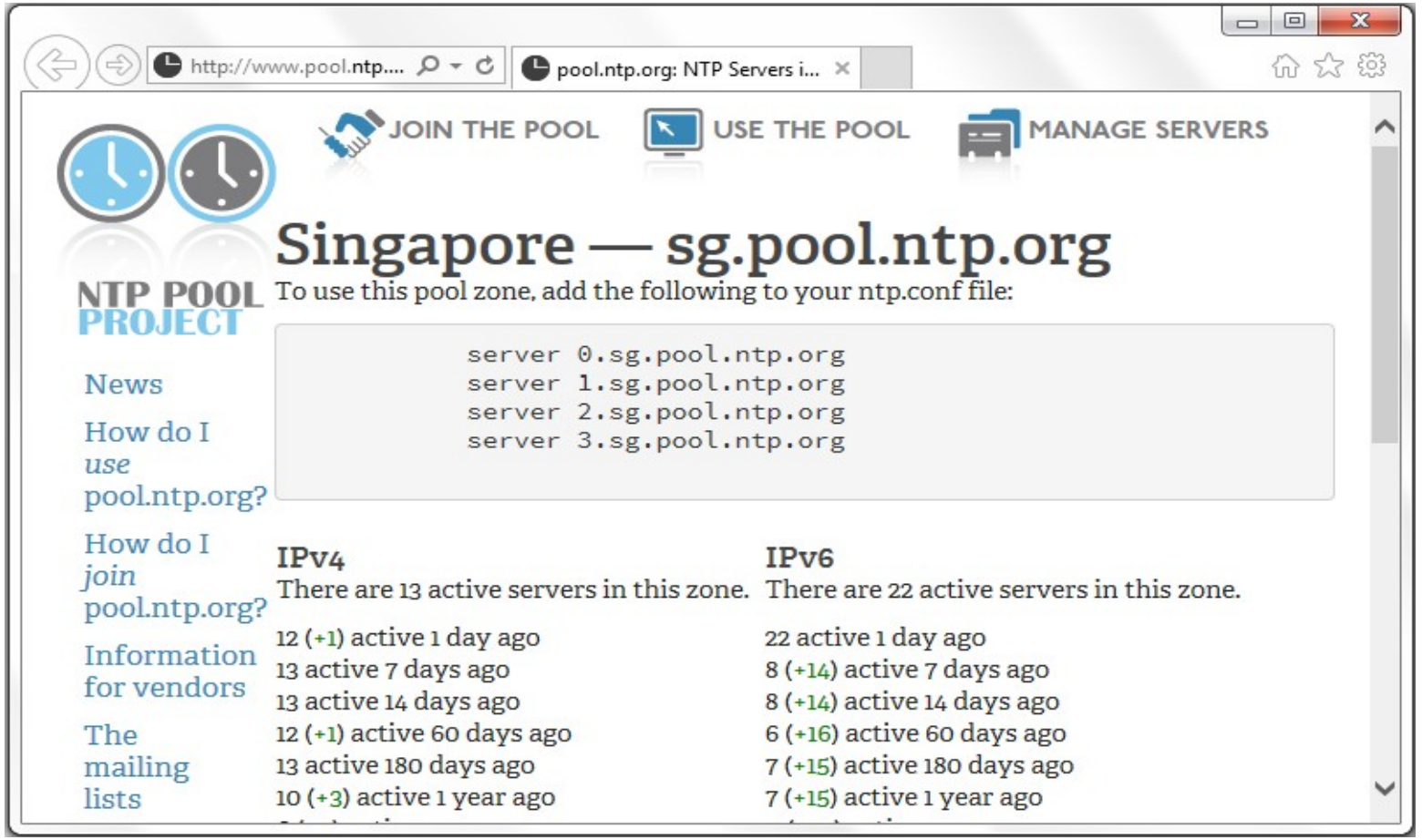
NTP uses the concept of stratum to describe how many NTP hops away a machine is from the most reliable authoritative time source at stratum 0.

As the stratum number increases, its accuracy degrades.



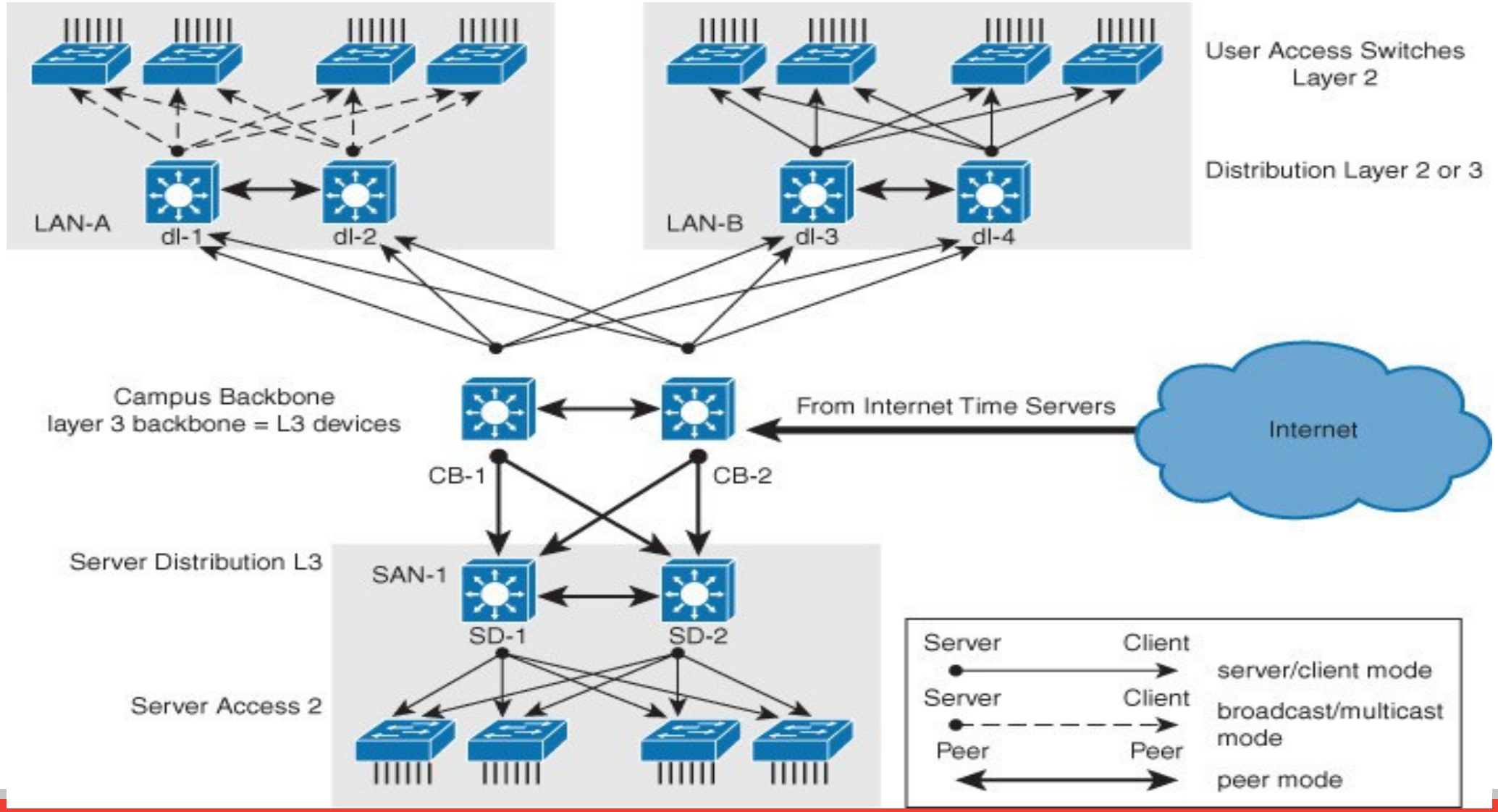
UTC (Coordinated Universal Time), aka GMT (Greenwich Mean Time), is the current worldwide standard for time and date.

An example is the NTP pool project which is maintaining a network of NTP servers that are being used by many systems around the world https://www.pool.ntp.org/zone/sg



To use the pool, simply configure your device to point to any of the provided domain name, each representing a random set of NTP servers.

After obtaining time from an NTP pool, an enterprise may implement time synchronization within its network in a hierarchical manner:



Step 1: Set a clock to current date/time

To be specific, we'll now discuss how to configure the network devices to synchronize time using NTP

Before configuring NTP, you may wish to configure your device clock to current date/time:

```
R1# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)# clock timezone SGT 8
R1(config)# ^Z
R1#
R1# clock set 20:52:00 25 October 2017
*Oct 25 12:52:00.000: %SYS-6-CLOCKUPDATE: System clock has been updated from 20:36:38
  SGT Wed Oct 25 2015 to 20:52:00 SGT Wed Oct 25 2015, configured from console by
  console.
R1# show clock
20:52:55.051 SGT Wed Oct 25 2017
```

Any letters representing your timezone, e.g. SGT for Singapore Time, which is UTC + 8 hrs.

Based on the design of your network, configure network devices as NTP server, client, peer or broadcast:

- Server: provides time information to clients, only used if you are not able to reach reliable external NTP servers

```
R1(config)# ntp master [stratum]
```

- Client: synchronizes time with an NTP server

```
R1(config)# ntp server {ip | hostname} [version] [prefer]
```

{ip/hostname of the NTP master R1}

- Peer: also called symmetric mode, which work both as server and client to synchronize, or be synchronized by each other

```
R1(config)# ntp peer {ip | hostname} [version] [prefer]
```

- Broadcast: pushes time announcements from NTP server to clients, used only when time accuracy is not a concern

Server to broadcast:

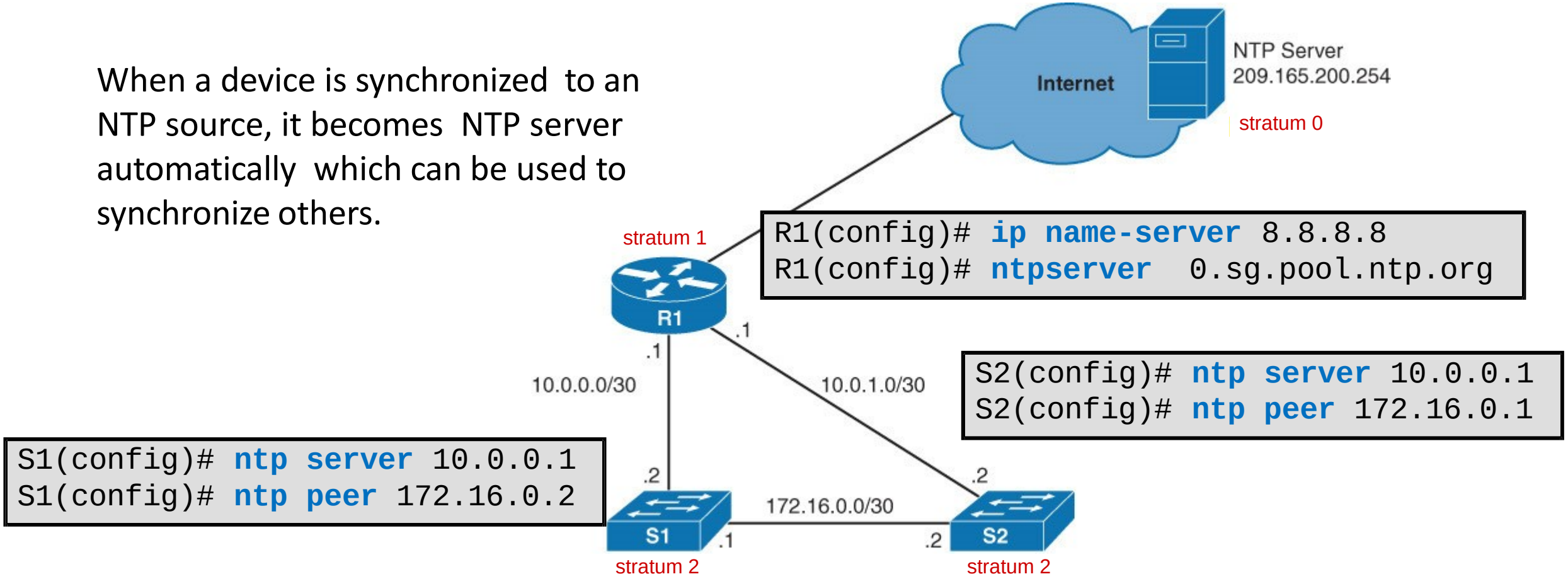
```
R1(config-if)# ntp broadcast [version]
```

Client to receive:

```
R1(config-if)# ntp broadcast client
```

An example of configuring NTP, with edge router R1 synchronizing with an external NTP time source, which in turn acts as NTP server for internal switches

When a device is synchronized to an NTP source, it becomes NTP server automatically which can be used to synchronize others.




```
R1(config)# ip name-server 8.8.8.8
R1(config)# ntpserver  0.sg.pool.ntp.org
```

```
S2(config)# ntp server 10.0.0.1
S2(config)# ntp peer 172.16.0.1
```

```
S1(config)# ntp server 10.0.0.1
S1(config)# ntp peer 172.16.0.2
```

To verify that the device's clock has indeed been synchronized with NTP, use the following show commands:

To verify which NTP server your device is synchronizing with, e.g.:

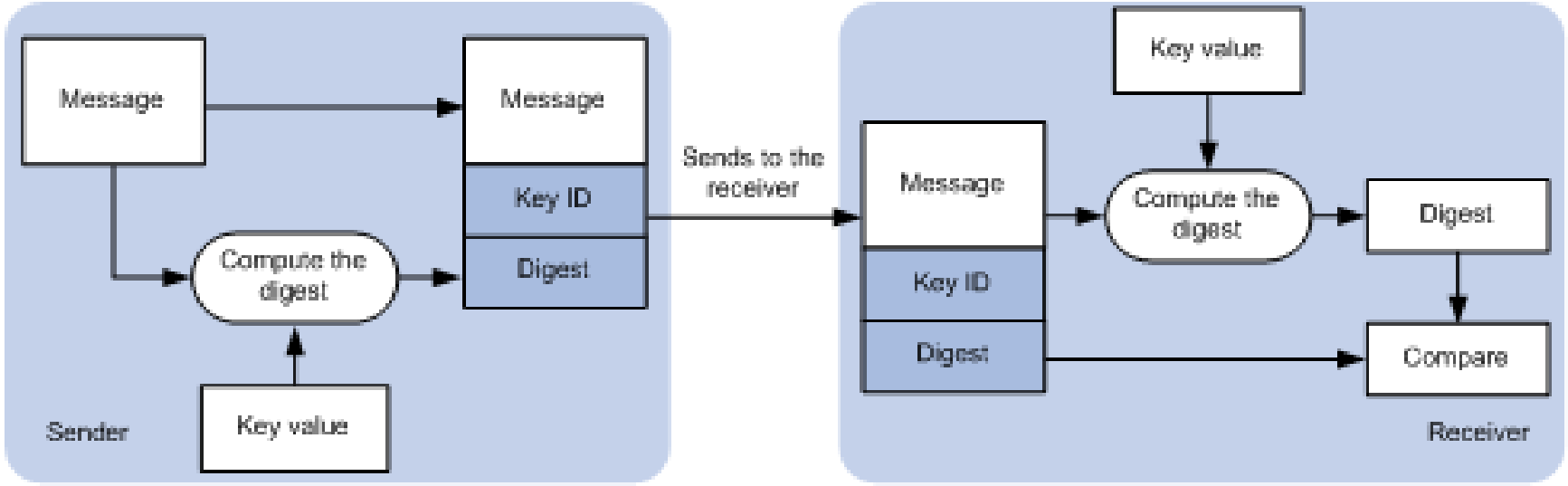
```
R1# show ntp associations

  address         ref clock       st   when   poll reach  delay  offset   disp
*~209.165.200.254 .LOCL.          1     24     64    17  1.000  -0.500  2.820
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

By synchronizing with stratum 1 time source, your device automatically becomes stratum 2 time source as shown, e.g.:

```
R1# show ntp status
Clock is synchronized, stratum 2, reference is 209.165.200.254
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**10
ntp uptime is 1500 (1/100 of seconds), resolution is 4000
reference time is D67E670B.0B020C68 (05:22:19.043 PST Mon Jan 13 2014)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 630.22 msec, peer dispersion is 189.47 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 64, last update was 5 sec ago.
```

Unfortunately, an attacker can inject false NTP replies to mess up time synchronization and log timings. Thus, it is advisable to enable NTP authentication to protect NTP clients

Both NTP client and server are configured with the same pre-shared key/password which is used to compute MD5 hash for authentication:



Using the pre-shared key, NTP server generates an MD5 hash and send it together with NTP message.

Using the same pre-shared key, NTP client re-computes the MD5 hash and verifies it with the received MD5 hash before accepting the NTP message.

For configuring of NTP authentication, the additional commands required are as follows:

- Define one or more NTP key(s), each number specifies a unique NTP key (password):

```
R1(config)# ntp authentication-key key-number md5 string
```

- Specify which key to enable for authentication:

```
R1(config)# ntp trusted-key key-number
```

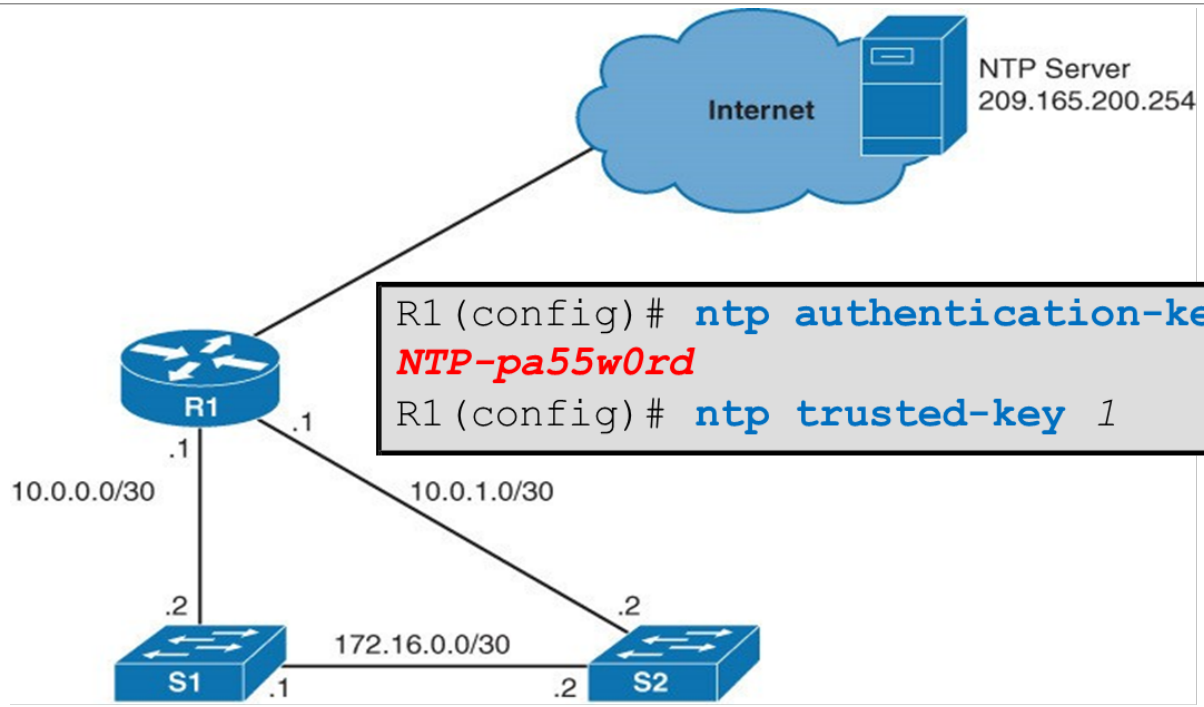
- Enable NTP authentication:

```
R1(config)# ntp authenticate
```

- For NTP client to specify which key to use for authentication:

```
R1(config)# ntp server {ip | hostname} key key-number
```

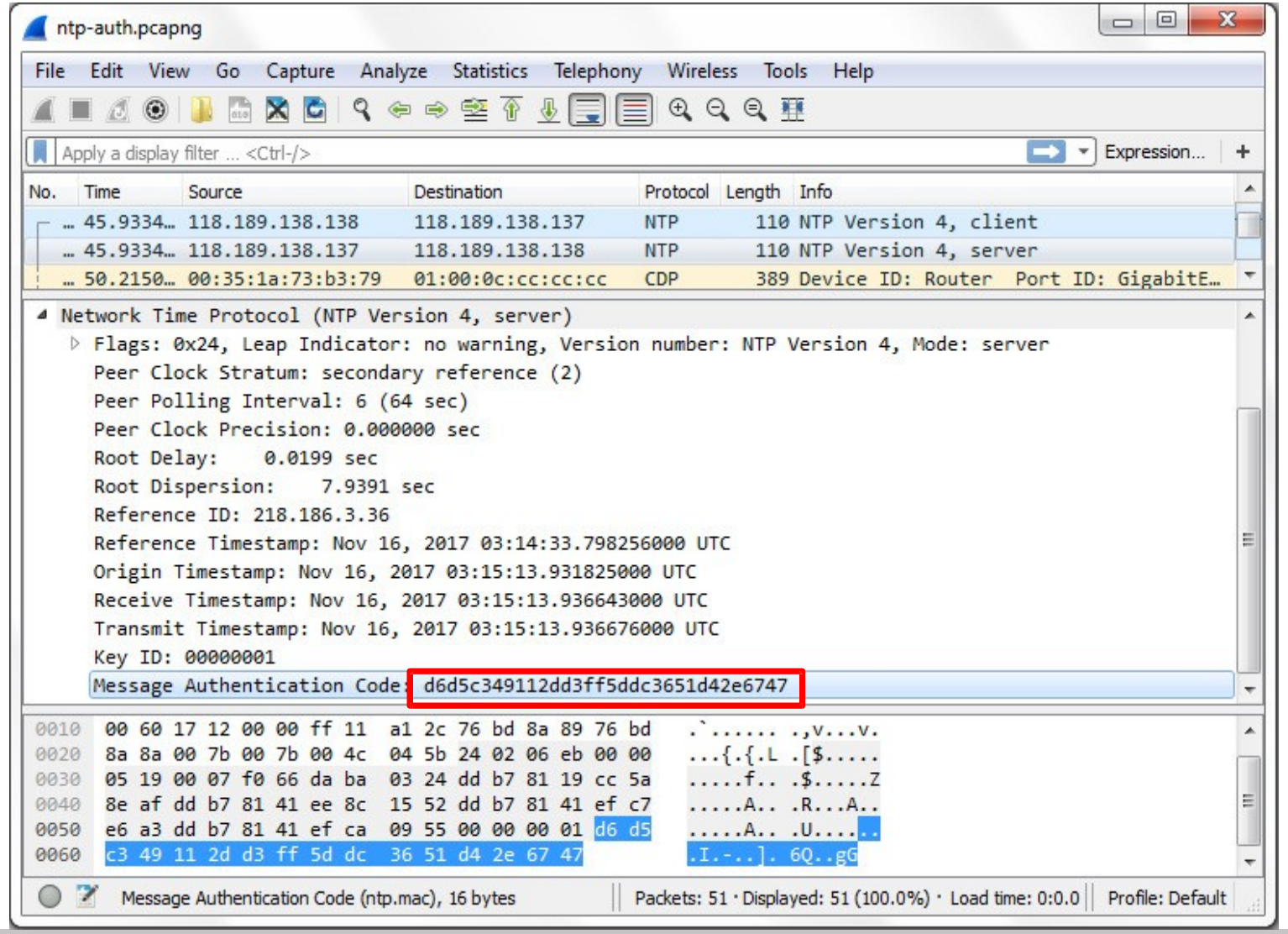
An example of configuring NTP authentication to prevent NTP clients from synchronizing with attacker's time source



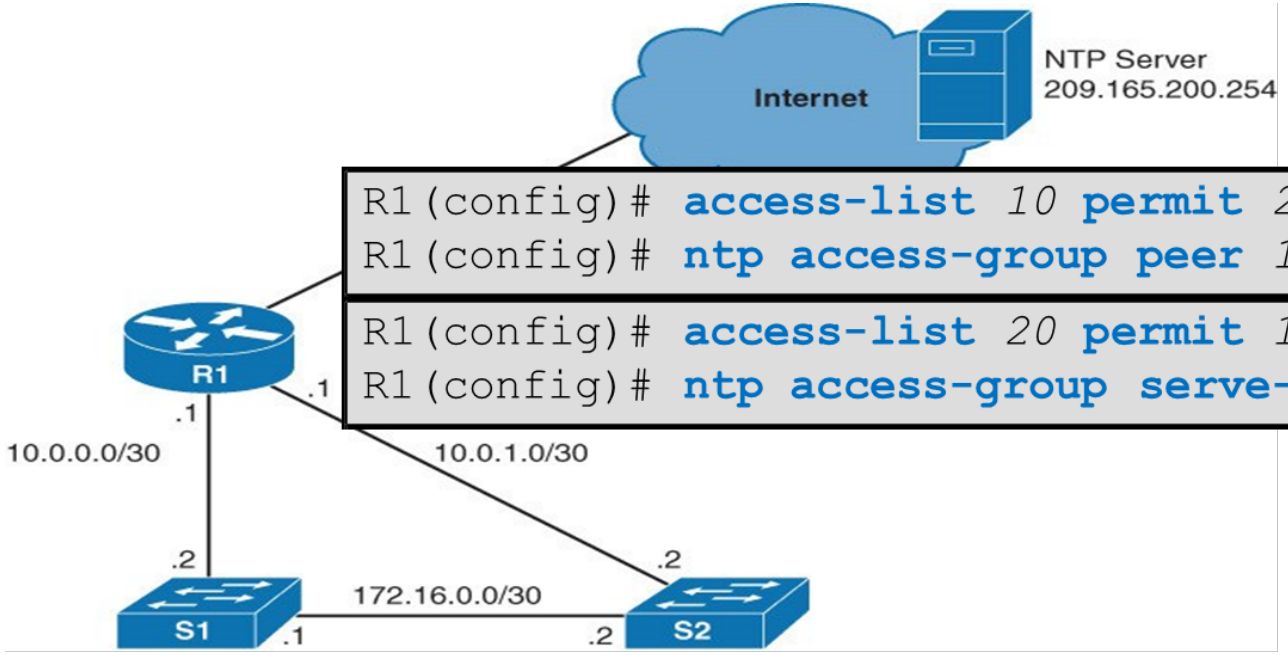

```
R1(config)# ntp authentication-key 1 md5
NTP-pa55w0rd
R1(config)# ntp trusted-key 1
```

```
S1(config)# ntp authentication-key 1 md5 NTP-pa55w0rd
S1(config)# ntp authenticate
S1(config)# ntp trusted-key 1
S1(config)# ntp server 10.0.0.1 key 1
```
{ to securely talk to the NTP server }

An example of NTP message containing MD5 hash (message authentication code) for authentication.



In addition, NTP can be protected by ACLs…




```
R1(config)# access-list 10 permit 209.165.200.254
R1(config)# ntp access-group peer 10
```

```
R1(config)# access-list 20 permit 10.0.0.0 0.0.0.7
R1(config)# ntp access-group serve-only 20
```

permit to allow NTP server to enter the network

Command for configuring ACL:

```
ntp access-group {query-only | serve-only | serve | peer}
access-list-number
```

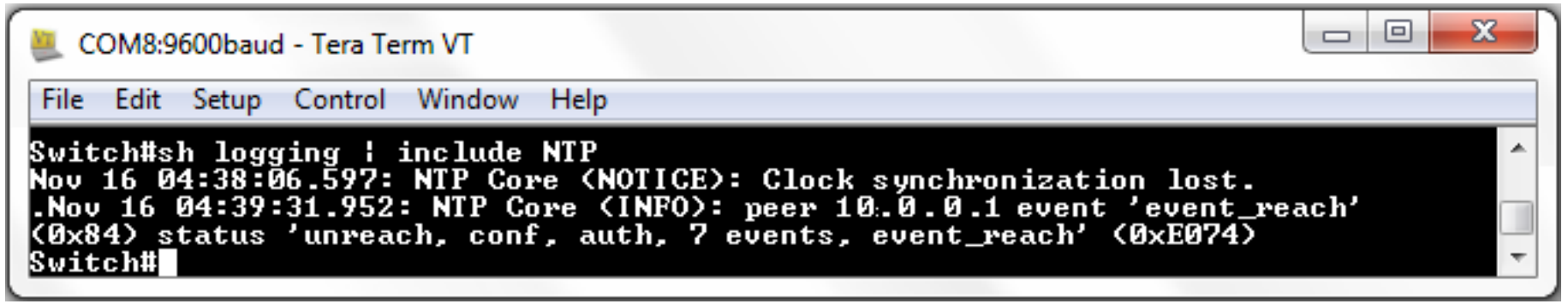
- peer: accept NTP reply and also respond to NTP request
- serve-only: for server to respond to NTP request only
- serve: similar as 'serve-only', but also accept NTP control queries
- query-only: only accept NTP control queries such as SNMP

Since NTP is important, it is also recommended to enable NTP logging to log significant NTP events.

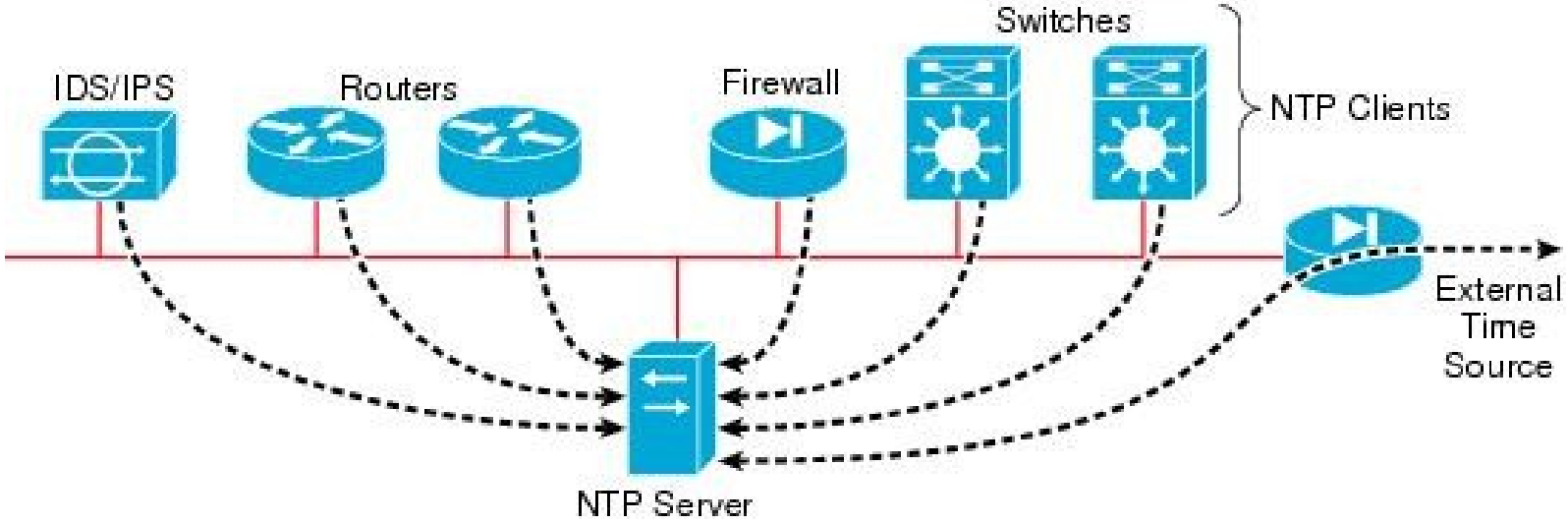
Enabling NTP logging (logging centrally will be discussed shortly):

```
Switch(config)# ntp logging
```

Examples of NTP logs:



If out-of-band (OOB) management network is available, it is more direct and secure to perform NTP time synchronization over OOB, e.g.:



If VRF is used: (note that command may be slightly different for different devices)

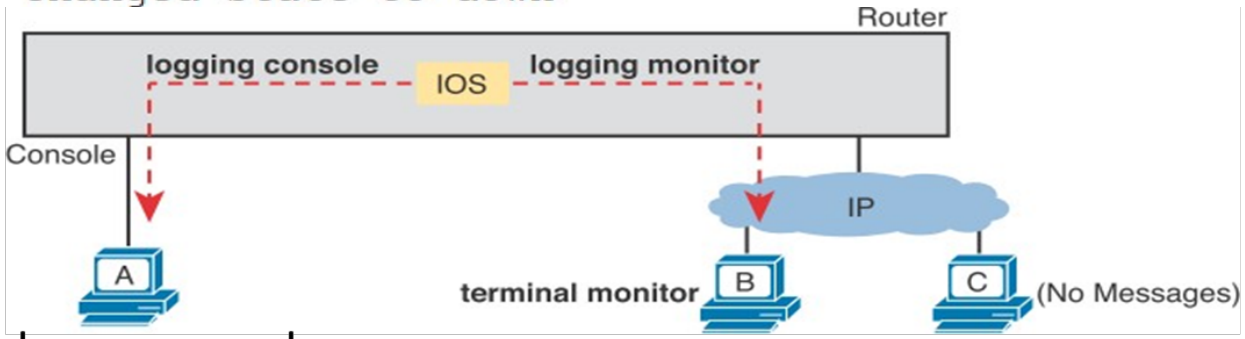
```
R1(config)# ntp server vrf Mgmt-vrf 10.0.0.1
```

```
S1(config)# ntp server 10.0.0.1 use-vrf Mgmt-vrf
```

(9.1 – *Part 3*) Syslog (System message logging) (RFC 5424) is a standard protocol for event notification messages which is widely adopted by the industry.

While configuring Cisco devices in the labs, you must have seen or even be annoyed by the syslog messages that appeared, e.g.

```
*Dec 18 17:10:15.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface
  FastEthernet0/0, changed state to down
```



Disable syslog on console:

```
R1(config)# no logging console
```

Prevent syslog from interrupting command entry:

```
R1(config)# logging synchronous
```

Enable syslog over Telnet/SSH:

```
R1(config)# logging monitor
```

In addition, at B:

```
R1# terminal monitor
```

For monitoring of network security, it is more useful to store log messages in the device RAM, or even better to a Centralized Syslog server.

By default, Syslog messages are transported over UDP port 514 (RFC 5426) or more securely over TLS over TCP port 6514 (RFC 5425) – may not be supported in Cisco.

To log messages into RAM:

```
R1(config)# logging buffered
```

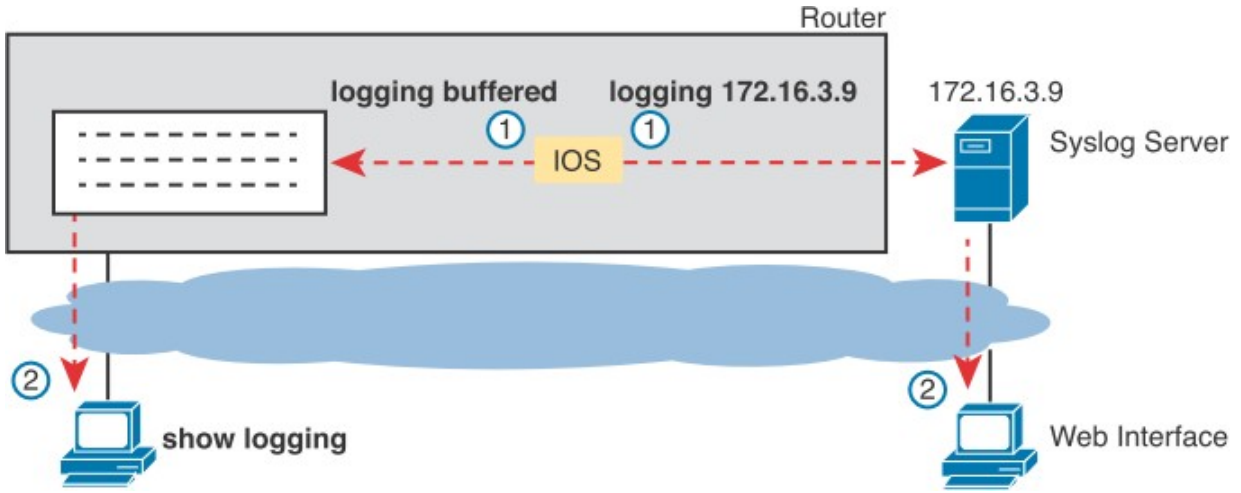
In addition:

```
R1# show logging
```

To log messages to Syslog server:

```
R1(config)# logging
172.16.3.9
R1(config)# logging trap 4
```

All errors 4 or below (more serious) are logged.

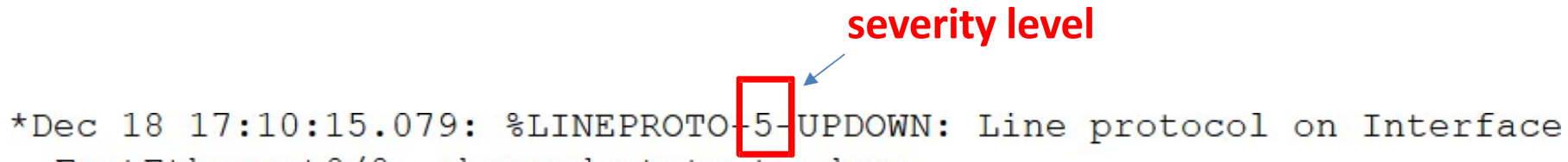


To prevent cluttering the log with too much data, Syslog has a severity level rating which you can use it to control which messages to log

Keyword	Numeral	Description	
Emergency	0	System unusable	Severe
Alert	1	Immediate action required	
Critical	2	Critical Event (Highest of 3)	Impactful
Error	3	Error Event (Middle of 3)	
Warning	4	Warning Event (Lowest of 3)	
Notification	5	Normal, More Important	Normal
Informational	6	Normal, Less Important	
Debug	7	Requested by User Debug	Debug

Service	To Set Message Levels
Console	logging console *level-name* | *level-number*
Monitor	logging monitor *level-name* | *level-number*
Buffered	logging buffered *level-name* | *level-number*
Syslog	logging trap *level-name* | *level-number*

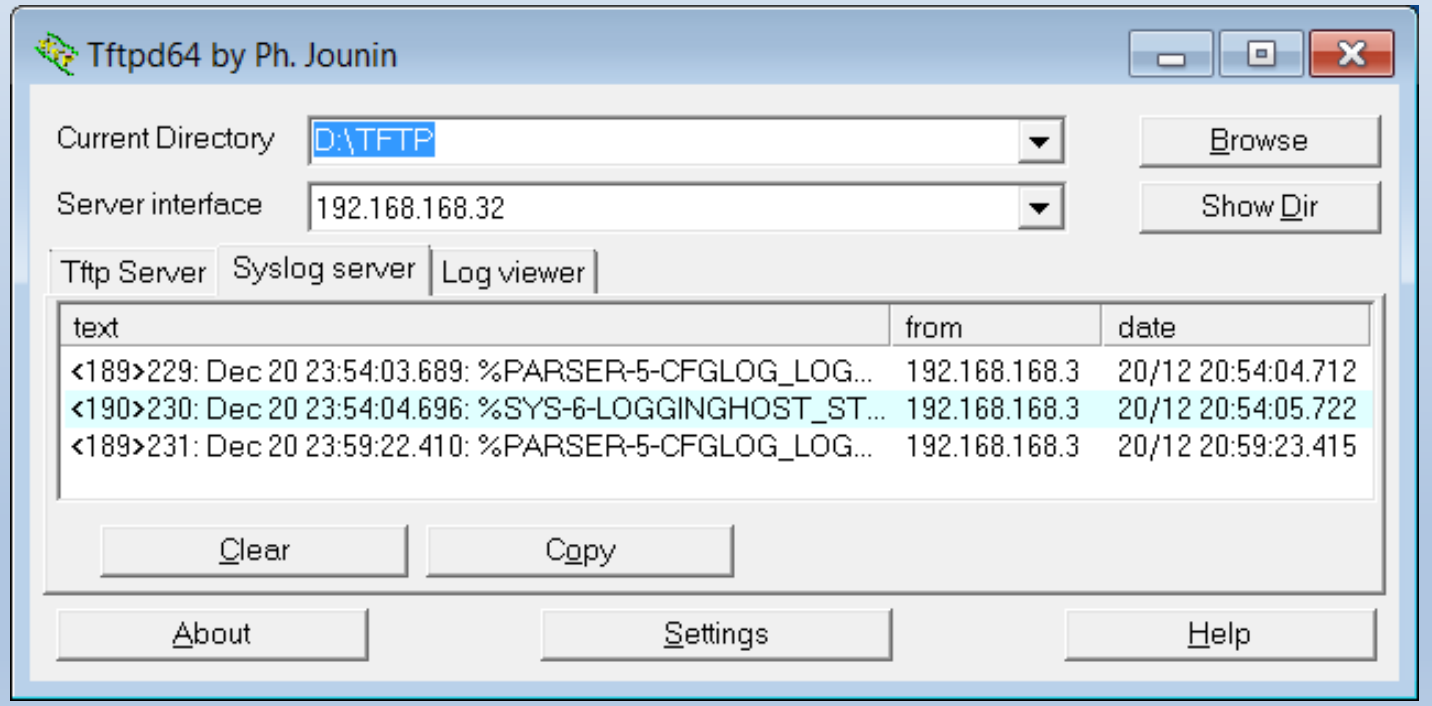
An example of Syslog message which shows the date/time of occurrence, the severity level and the description.



- **A timestamp:** *Dec 18 17:10:15.079
- **The facility on the router that generated the message:** %LINEPROTO
- **The severity level:** 5
- **A mnemonic for the message:** UPDOWN
- **The description of the message:** Line protocol on Interface FastEthernet0/0, changed state to down

There are many syslog servers available. An example is the free open source tftpd server which can also function as syslog server that you can try in the lab

http://tftpd32.jounin.net/



Alternatively, you may also try other Syslog servers like Kiwi Syslog which offers a free edition supporting up to 5 clients.

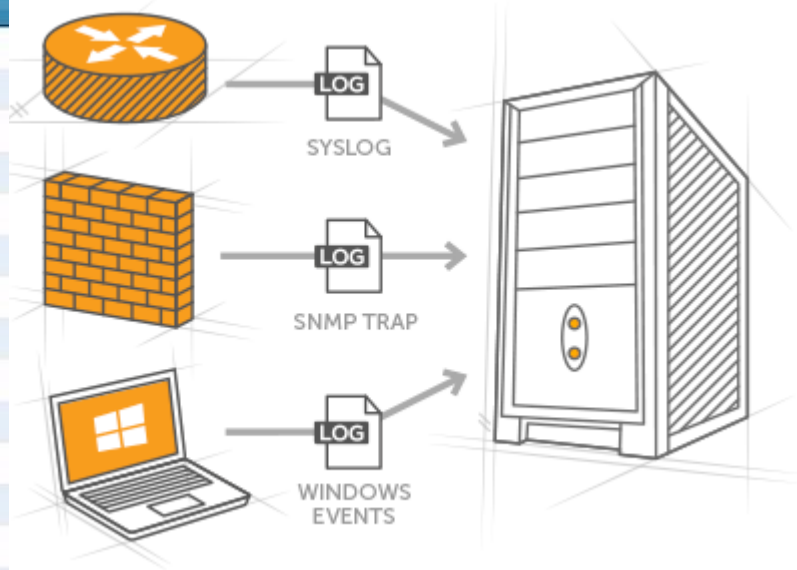
https://www.kiwisyslog.com/free-tools/kiwi-free-syslog-server

KIWI SYSLOG WEB ACCESS v1.5.1

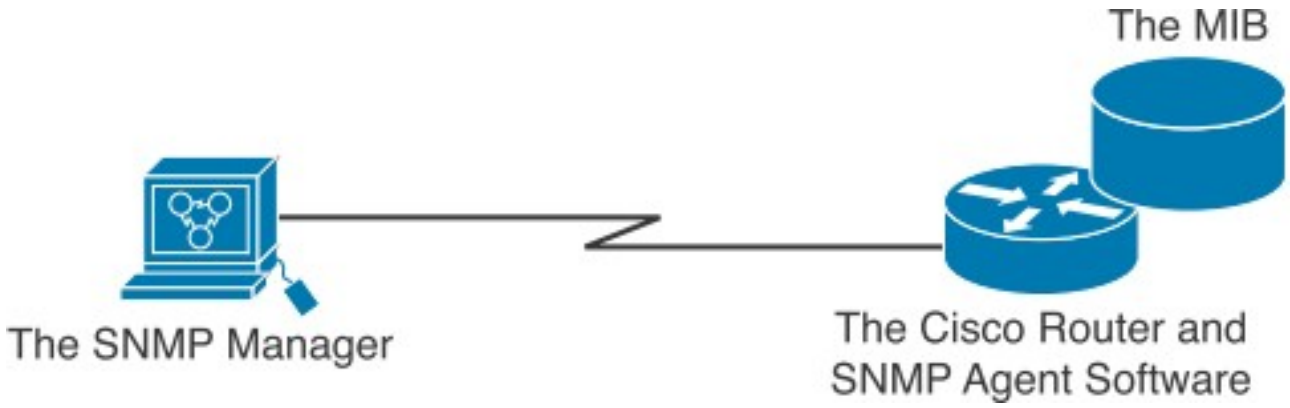
Events | Filters | Highlighting | Settings | Admin

Filter: [All Kiwi Syslog Server Events]

Date	Time	Facility	Level	Host Name	Message Text
2015-03-20	14:05:00	Cron	Warning	10.111.201.39	Logon failed
2015-03-20	14:05:00	Local1	Warning	10.111.201.39	Logoff Successful
2015-03-20	14:05:00	Local7	Critical	10.111.201.39	This is a test message generated by Kiwi SyslogGen
2015-03-20	14:05:00	Local1	Alert	10.111.201.39	Logon successful
2015-03-20	14:04:59	Cron	Error	10.111.201.39	Logon failed
2015-03-20	14:04:59	System1	Critical	10.111.201.39	Logoff Successful
2015-03-20	14:04:59	Syslog	Error	10.111.201.39	This is a test message generated by Kiwi SyslogGen
2015-03-20	14:04:59	System0	Critical	10.111.201.39	Logon successful
2015-03-20	14:04:58	Cron	Warning	10.111.201.39	Logon failed
2015-03-20	14:04:58	System2	Info	10.111.201.39	Logoff Successful
2015-03-20	14:04:58	User	Notice	10.111.201.39	This is a test message generated by Kiwi SyslogGen
2015-03-20	14:04:58	News	Debug	10.111.201.39	Logon successful
2015-03-20	14:04:57	Cron	Info	10.111.201.39	Logon failed



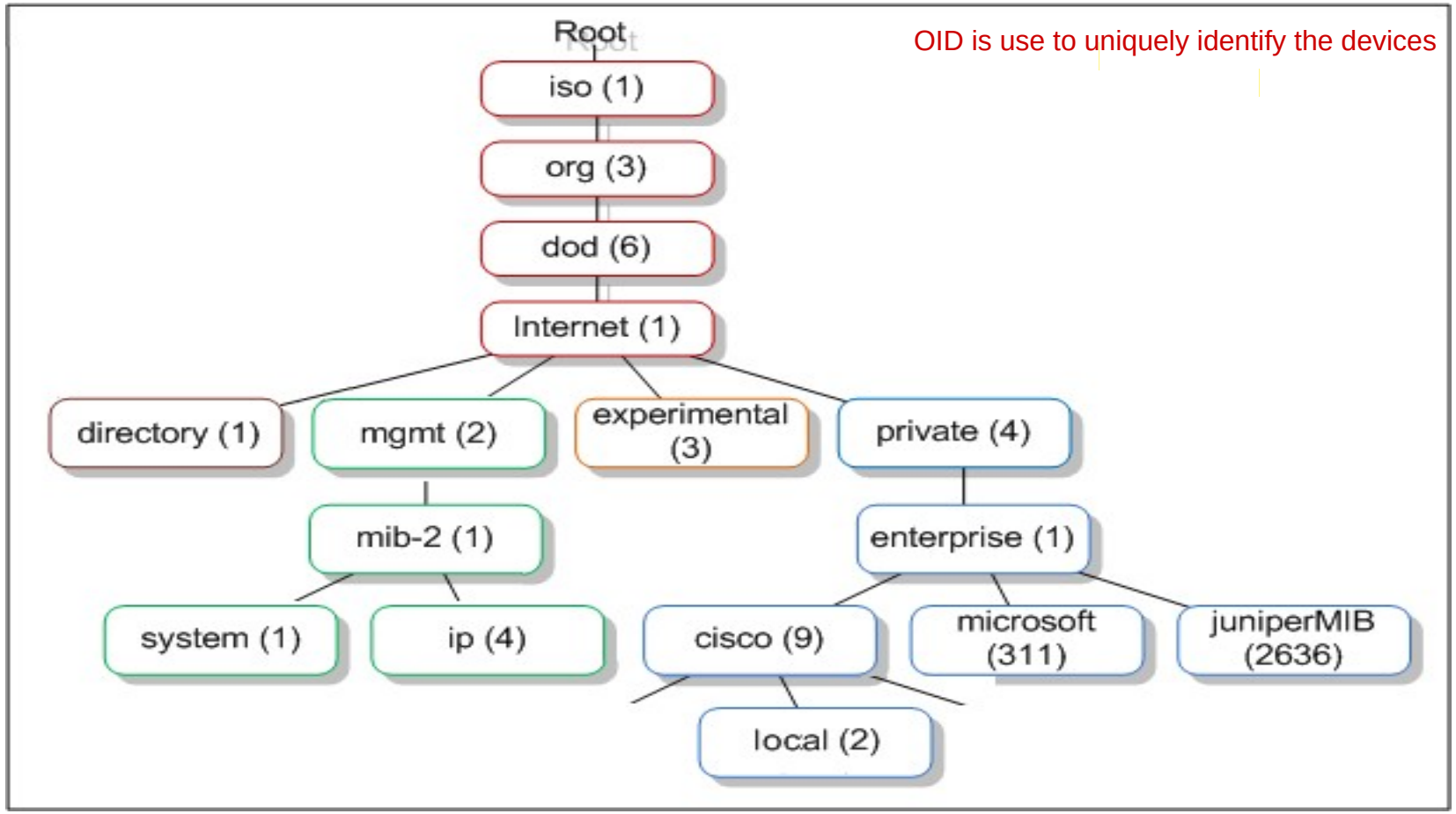
(9.2) SNMP is another protocol for event notification messages. In fact, SNMP is also designed for managing network devices, hence the name Simple Network Management Protocol.



There are 3 main components in a SNMP system:

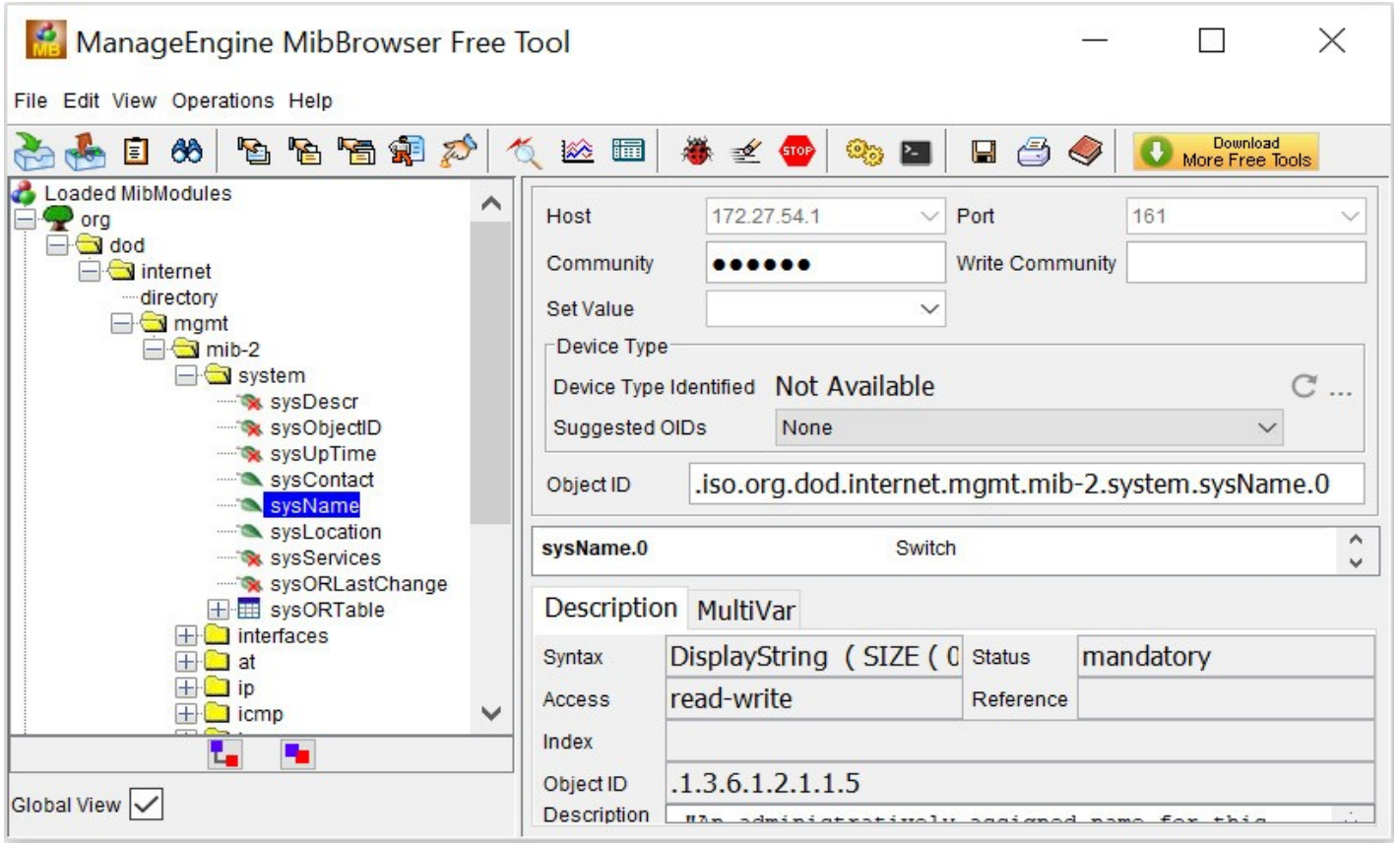
- SNMP manager – aka NMS (Network Management Station) which runs the network management application
- SNMP agent – running on the device to be managed (e.g. router) to read or write to the variables in the MIB
- MIB (Management Information Base) – a collection of managed objects (variables) and their associated values in the devices

The MIB is organized as a tree structure where the managed objects (variables) are at the leaf nodes, each identified by a long sequence of dotted numbers called OID (Object ID)



An example of NMS is the free ManageEngine MibBrowser where you can see the MIB tree of the device being managed

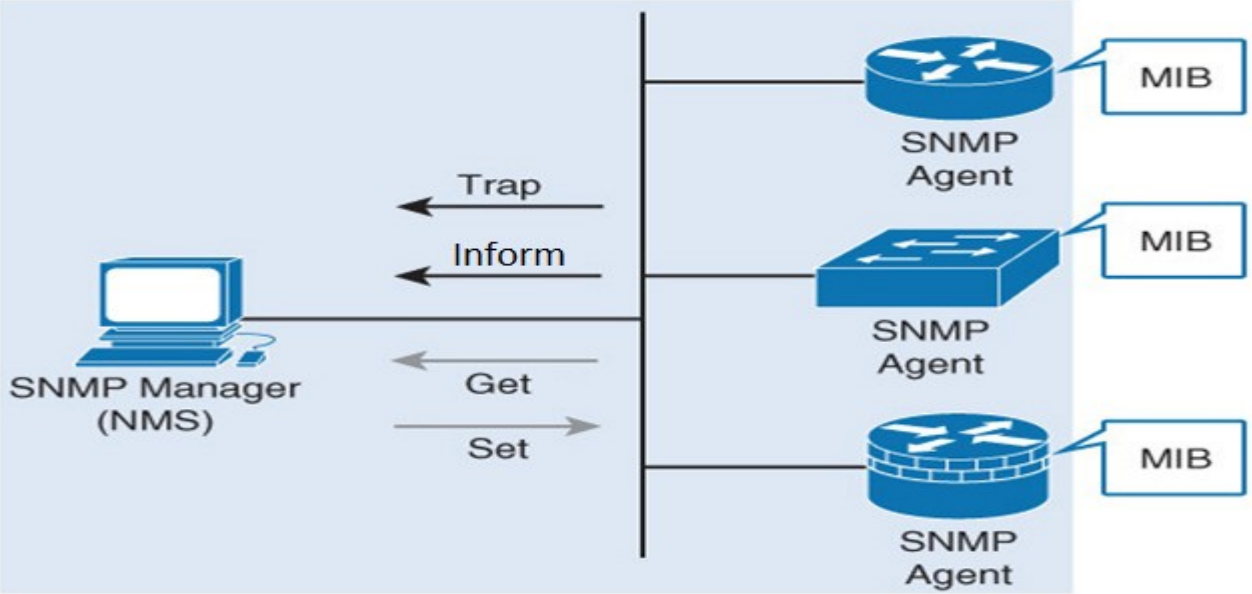
https://www.manageengine.com/products/mibbrowser-free-tool/download.html



The SNMP agents communicate with NMS (SNMP manager) over UDP port 162 for event notifications and UDP port 161 for device management.

Types of SNMP event notifications:

- Trap: no acknowledgement required and hence can be lost
- Inform: agent will resend if no acknowledgement from NMS



Types of SNMP device management:

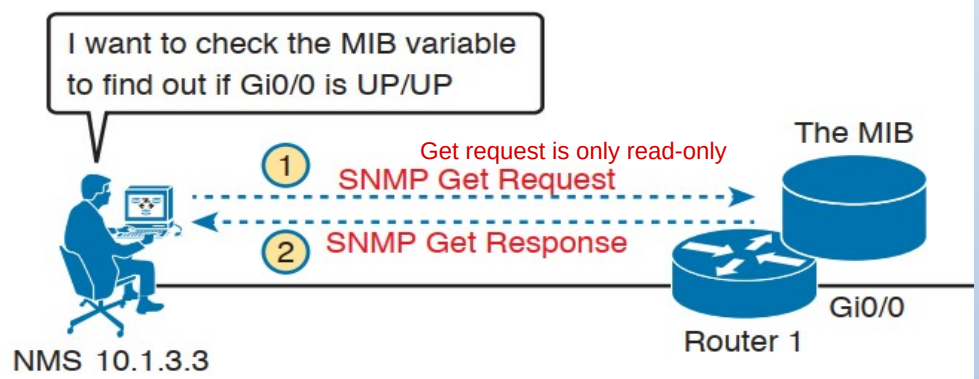
- Get: for NMS to have read-only (ro) access to device's MIB
- Set: for NMS to have read-write (rw) access to device's MIB

Today, three versions of SNMP exist. In this module, we'll discuss both SNMPv2c which is still commonly used and SNMPv3 which is more secure and should be used

Community string used in SNMPv1 and v2 is basically password, but is sent in plain text without encryption, and hence insecure!

SNMP Version	Security Level	Authentication	Encryption
SNMPv1	noAuthNoPriv	Community string	No
SNMPv2c	noAuthNoPriv	Community string	No
SNMPv3	noAuthNoPriv	Username	No
	authNoPriv	MD5 or SHA-1	No
	authPriv	MD5 or SHA-1	DES, 3DES, or AES

To enable SNMPv2 in network devices to be remotely managed by NMS, configure community as follows:



To allow SNMP Get, configure ro access to devices; and advisable to configure ACL to only allow specific NMS for more security:

```
R1(config)# ip access-list standard ACL_SNMP
R1(config-std-nacl)# permit host 10.1.3.3 (Permit SNMP IP Manager Address)
R1(config)# snmp-server community secretROpw ro ACL_SNMP
```

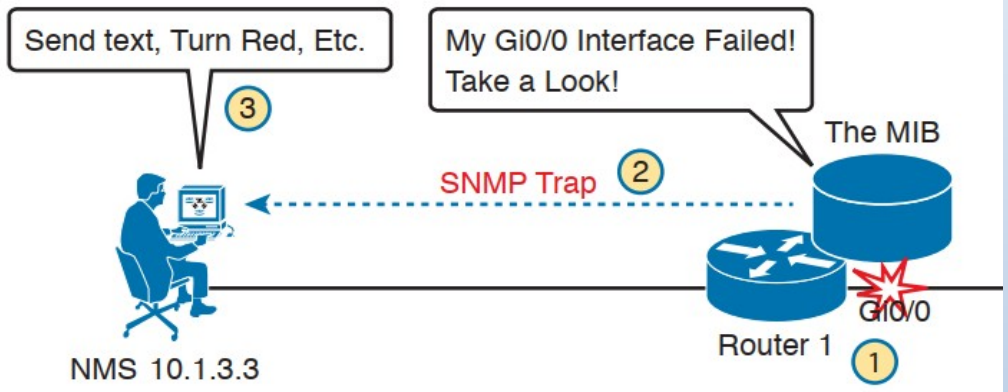
community string

To allow SNMP Set, configure rw access to devices:

```
R1(config)# snmp-server community secretRWpw rw ACL_SNMP
```

Mode of access

To enable SNMPv2 in network devices to send event notifications to NMS, configure as follows:



Community String

To send traps:

```
R1(config)# snmp-server enable traps
R1(config)# snmp-server host 10.1.3.3 version 2c secretTRAPpw
```

Alternatively, to send informs:

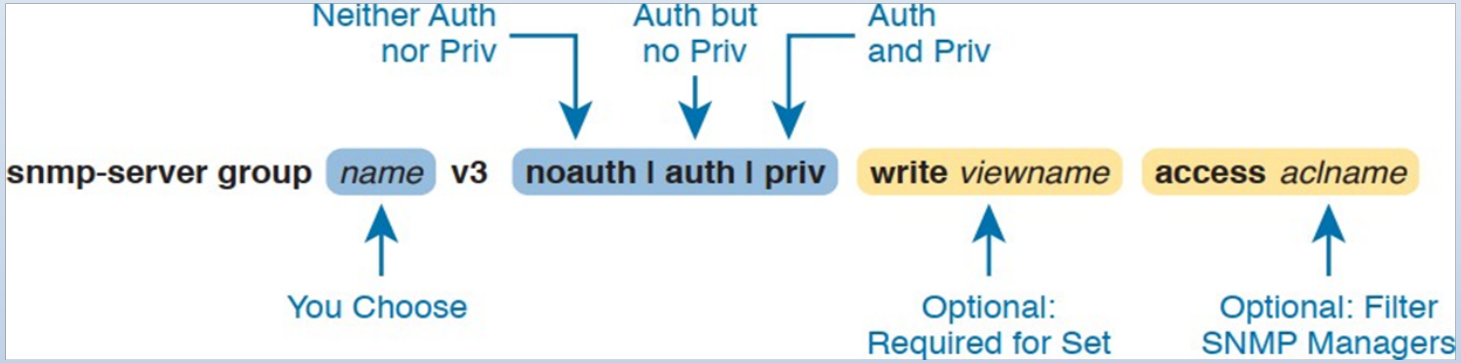
```
R1(config)# snmp-server enable traps
R1(config)# snmp-server host 10.1.3.3 informs version 2c
secretTRAPpw
```

To verify SNMPv2 configurations, use the show snmp command as follows:

```
R1# show snmp community

Community name: secretROpw
Community Index: secretROpw
Community SecurityName: secretROpw
storage-type: nonvolatile        active access-list: ACL_SNMP

Community name: secretRWpw
Community Index: secretRWpw
Community SecurityName: secretRWpw
storage-type: nonvolatile        active access-list: ACL_SNMP

Community name: secretTRAPpw
Community Index: secretTRAPpw
Community SecurityName: secretTRAPpw
storage-type: nonvolatile        active

R1# show snmp host
Notification host: 10.1.3.3      udp-port: 162    type: trap
user: secretTRAPpw       security model: v2c
```

To enable SNMPv3 for device management, the community string in SNMPv2 is replaced with the configuration of (1) SNMPv3 group, and (2) SNMPv3 username and password

(Step 1) Configuring SNMPv3 group:



E.g.: Configure SNMP *Group1* for SNMP GET:

```
R1(config)# ip access-list standard ACL_SNMP
R1(config-std-nacl)# permit host 10.1.3.3
R1(config)# snmp-server group Group1 v3 auth read viewname
access ACL_SNMP
```

Configure SNMP *Group2* for SNMP SET:

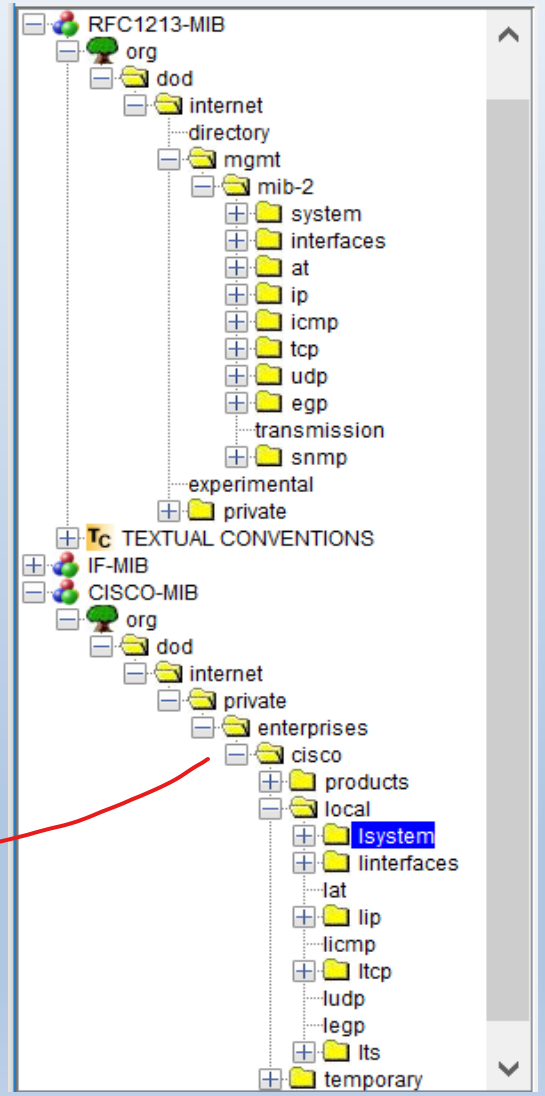
encypting

```
R1(config)# snmp-server group Group2 v3 priv write viewname
access ACL_SNMP
```

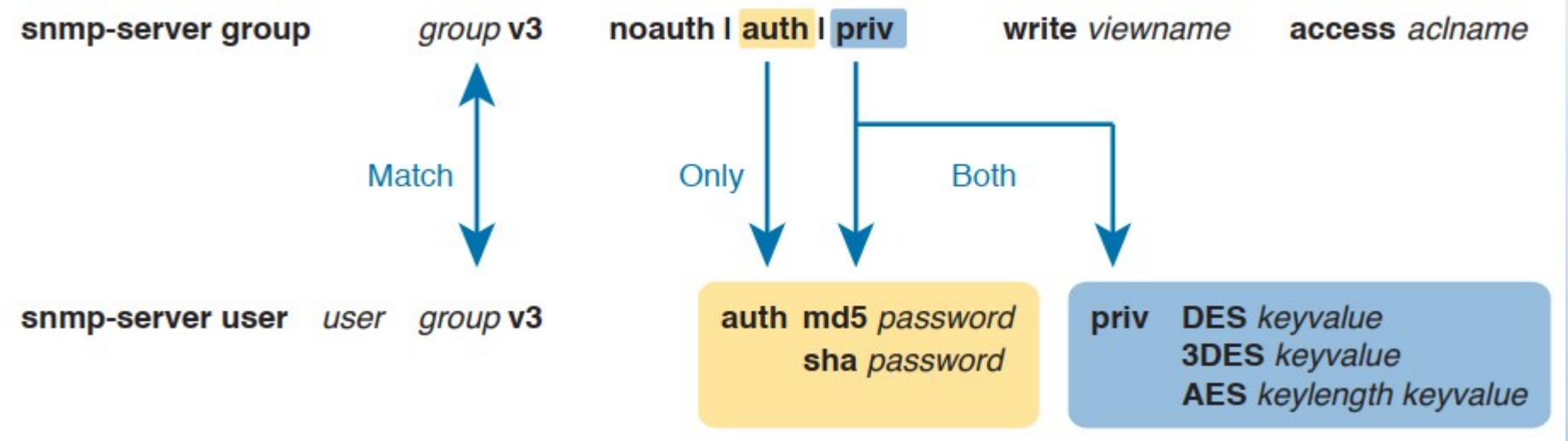
For additional security, SNMPv3 allows the configuration of view to restrict access to a subset of the MIB tree that the remote NMS can read from and/or write to the device.

An example of configuring view to restrict access:

```
R1(config)# snmp-server view viewname
mib-2 included
R1(config)# snmp-server view viewname
at excluded
R1(config)# snmp-server view viewname
cisco included
```



(Step 2) Configure SNMPv3 username and password based on the configured SNMPv3 group as follows:



Configure username and password in SNMP *Group1* for SNMP GET:

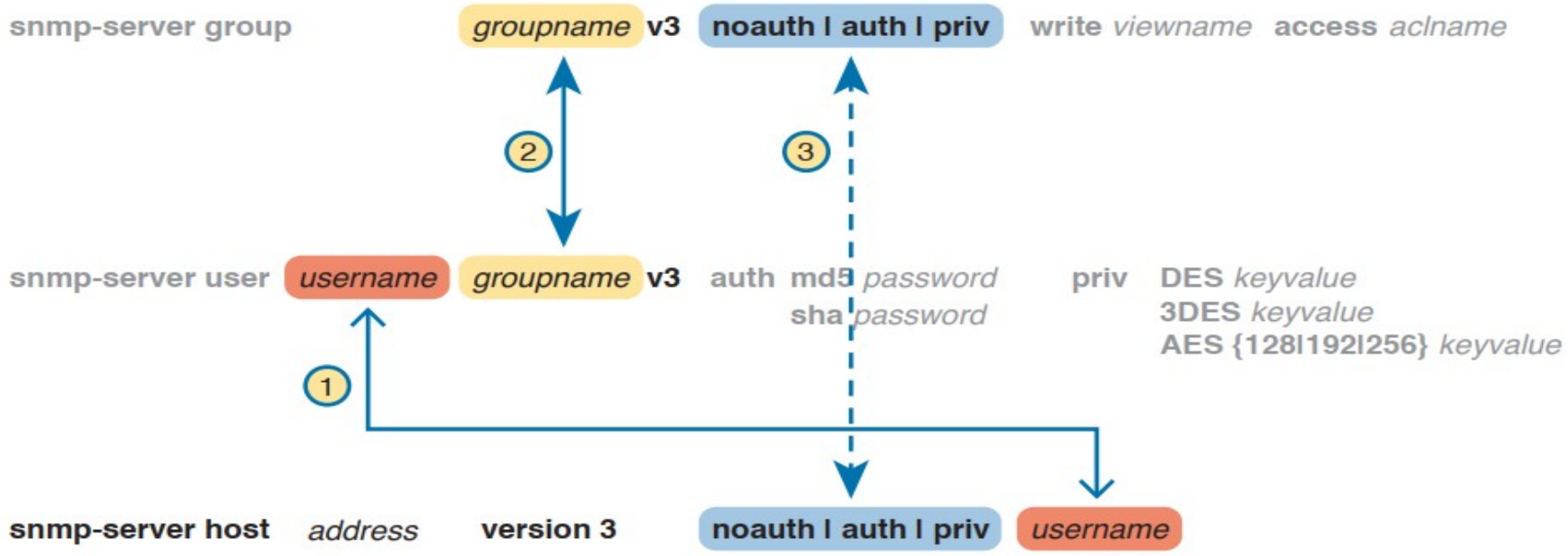
```
R1(config)# snmp-server user You1 Group1 v3 auth md5 cisco1
```

Configure username and password in SNMP *Group2* for SNMP SET:

```
R1(config)# snmp-server user You2 Group2 v3 auth sha cisco2
priv aes 128 cisco3
```

key for AES encryption

With the configuration of SNMPv3 group, username and password, network devices can also be enabled to use SNMPv3 to send event notifications using traps or informs to NMS.



To send traps:

```
R1(config)# snmp-server enable traps
R1(config)# snmp-server host 10.1.3.3 version 3 auth You1
```

To send informs:

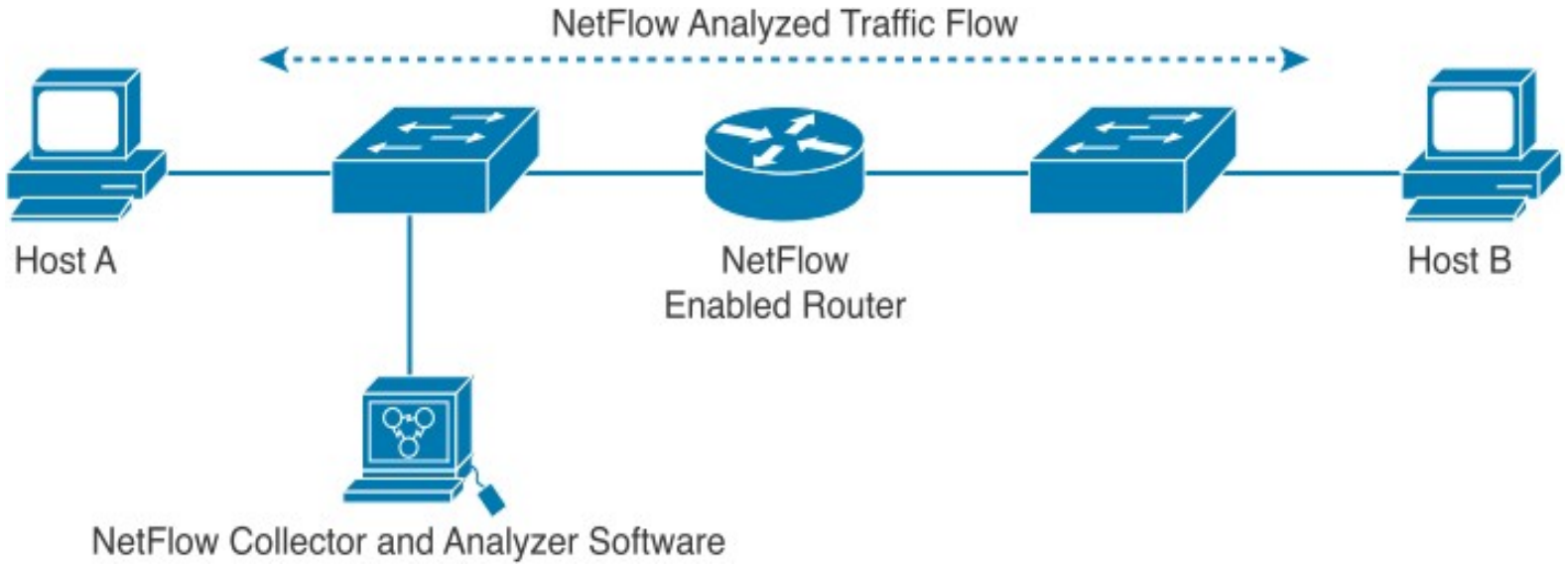
```
R1(config)# snmp-server host 10.1.3.3 informs version 3
priv You2
```

Similarly, to verify SNMPv3 configurations, use the show snmp command as follows:

```
R1# show snmp group
groupname: Group2                          security model:v3 priv
contextname: <no context specified>        storage-type: nonvolatile
readview : v1default                       writeview: v1default
notifyview: <no notifyview specified>
row status: active      access-list: ACL_SNMP

R1# show snmp user
User name: You2
Engine ID: 800000090300D48CB57D8200
storage-type: nonvolatile        active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: Group2

R3# show snmp host
Notification host: 10.1.3.3     udp-port: 162   type: inform
user: You2      security model: v3 priv
```

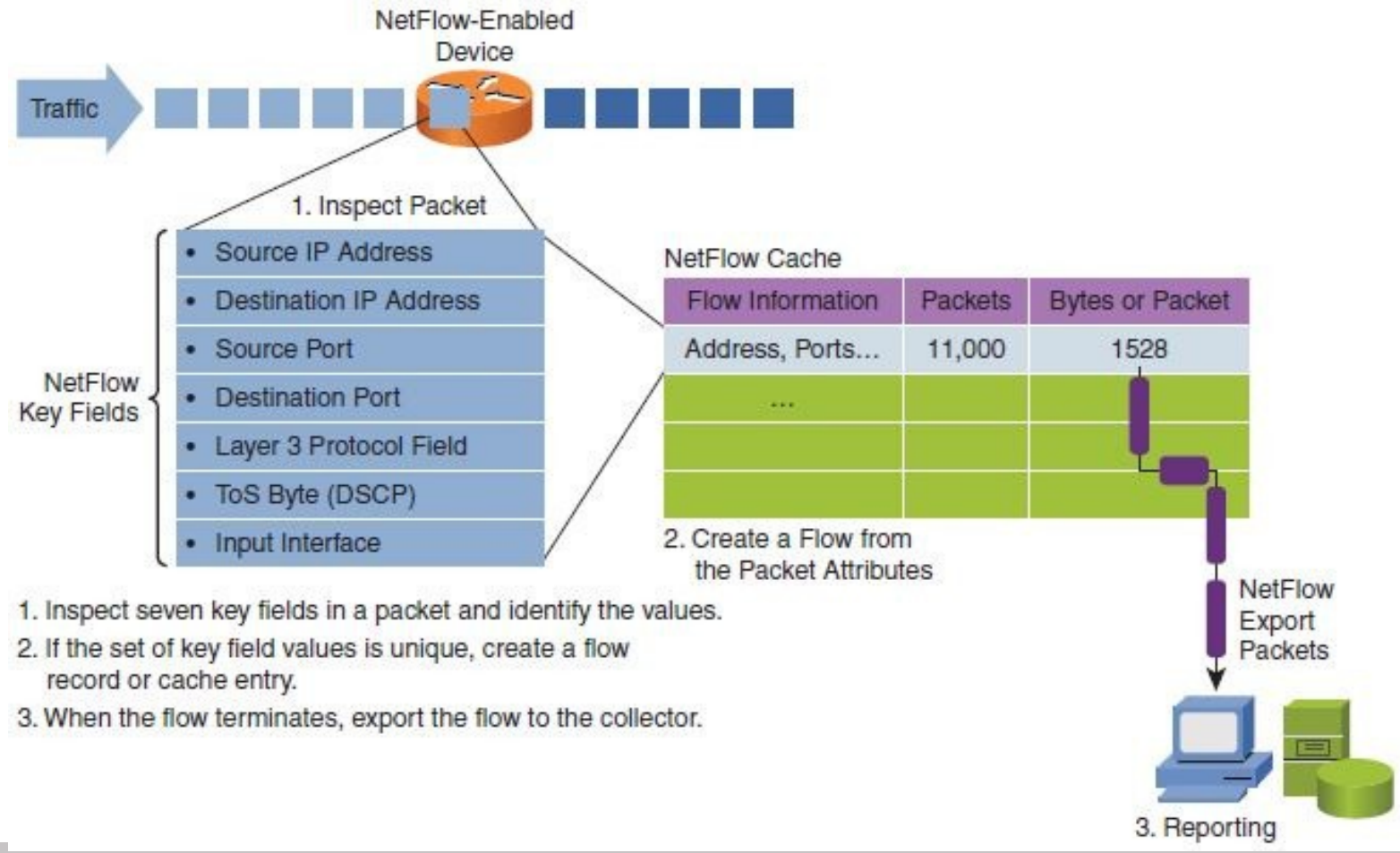
(9.3) To support network traffic monitoring, Cisco has developed NetFlow (RFC 3954) to collect statistics on IP packets flowing through network devices

When enabled, NetFlow keeps statistics of different IP packets flowing through a router in a NetFlow cache, which can then be exported to a NetFlow Collector for centralized logging and analysis.

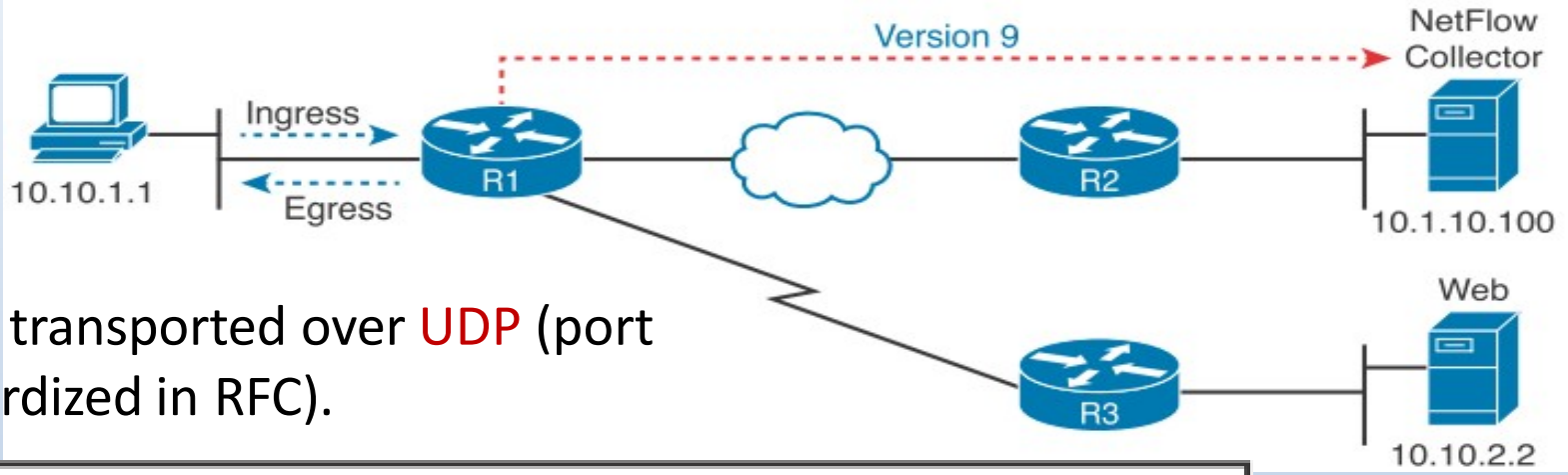


NetFlow has proven valuable and today different variants have been implemented, e.g. IPFIX (RFC 7011, derived from NetFlow), sFlow (RFC 3176) and JFlow (Juniper).

A flow is defined as a unidirectional sequence of packets that pass through a network device having the same values for certain key fields in the packet headers, examples:



The simplest way to enable NetFlow is to rely on default matching rules and thus only need to configure the following:



NetFlow is transported over UDP (port not standardized in RFC).

```
R1(config)# interface fastethernet0/0
R1(config-if)# ip flow ingress
R1(config-if)# ip flow egress
R1(config-if)# exit
R1(config)# ip flow-export destination 10.1.10.100 99
R1(config)# ip flow-export version 9
R1(config)# ip flow-export source loopback 0
R1(config)# end
```

Monitor incoming packets

Monitor outgoing packets

Port number

current version

For easier identification of router, use loopback 0 IP address to send NetFlow packets to Collector.

To verify NetFlow configuration, use the following commands:

```
R1# show ip flow interface
FastEthernet0/0
  ip flow ingress
  ip flow egress

R1# show ip flow export
Flow export v9 is enabled for main cache
  Export source and destination details :
  VRF ID : Default
    Source(1)        1.1.1.1 (Loopback0)
    Destination(1)   10.1.10.100 (99)
  Version 9 flow records
  0 flows exported in 0 udp datagrams
  0 flows failed due to lack of export packet
  0 export packets were sent up to process level
  0 export packets were dropped due to no fib
  0 export packets were dropped due to adjacency issues
  0 export packets were dropped due to fragmentation failures
  0 export packets were dropped due to encapsulation fixup failures
R1#
```

To view statistics stored locally in the NetFlow cache, use the command:

```
R1# show ip cache flow
IP packet size distribution (5727 total packets):
   1-32   64   96  128  160  192  224  256  288  320  352  384  416  448  480
   .000 .147 .018 .700 .000 .001 .001 .001 .001 .011 .009 .001 .002 .000 .001

    512  544  576 1024 1536 2048 2560 3072 3584 4096 4608
   .001 .001 .097 .000 .000 .000 .000 .000 .000 .000 .000

  .
  .
  .
Protocol          Total    Flows   Packets Bytes  Packets Active(Sec) Idle(Sec)
--------          Flows     /Sec     /Flow  /Pkt     /Sec      /Flow     /Flow
TCP-Telnet            4      0.0        27    43      0.2        5.0      15.7
TCP-WWW             104      0.2        14   275      3.4        2.1       1.5
ICMP                  4      0.0      1000   100      8.8       27.9      15.4

SrcIf         SrcIPaddress    DstIf         DstIPaddress    Pr SrcP DstP  Pkts
  .
  .
  .
```

Instead of relying on default matching rules, NetFlow version 9 allows the flexibility to configure matching rules and collection statistics, hence also called flexible NetFlow

(1) Example of configuring matching rules and collection statistics using flow record:

```
R1(config)# flow record my-record
R1(config-flow-record)# match ipv4 source address
R1(config-flow-record)# match ipv4 destination address
R1(config-flow-record)# match transport source-port
R1(config-flow-record)# match transport destination-port
R1(config-flow-record)# match ipv4 protocol
R1(config-flow-record)# match ipv4 tos
R1(config-flow-record)# match interface input
R1(config-flow-record)# collect counter bytes
R1(config-flow-record)# collect counter packets
```

Next, apply the specific matching rules and collection statistics configured onto the interface of the router as follows:

(2) Configuring flow exporter:

```
R1(config)# flow exporter my-exporter
R1(config-flow-exporter)# destination 10.1.10.100
R1(config-flow-exporter)# transport udp 99
R1(config-flow-exporter)# source lo0
R1(config-flow-exporter)# export-protocol netflow-v9
```

(3) Configuring flow monitor:

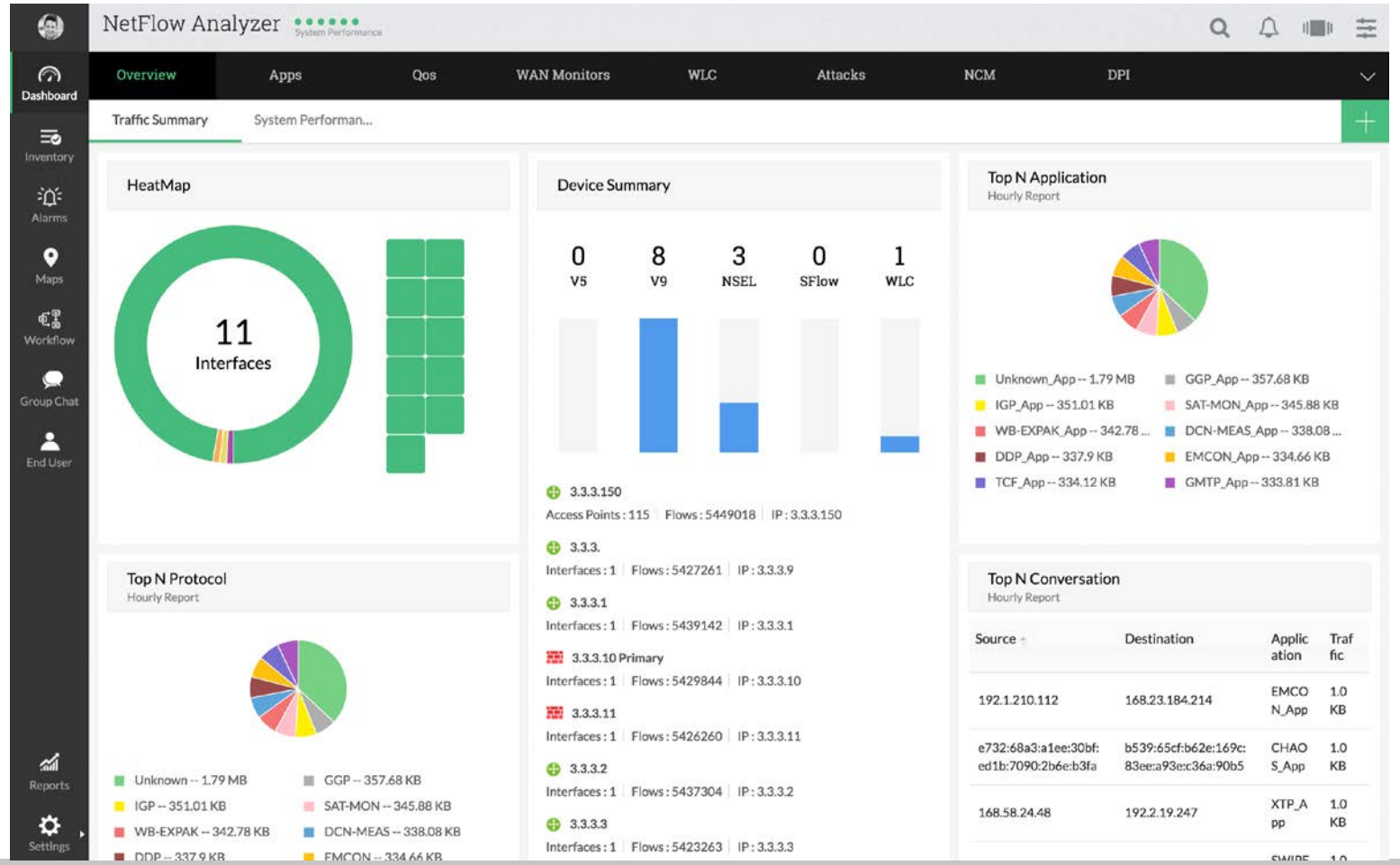
```
R1(config)# flow monitor my-monitor
R1(config-flow-monitor)# exporter my-exporter
R1(config-flow-monitor)# record my-record
```

(4) Applying flow monitor onto router interface:

```
R1(config)# interface g0/0
R1(config-if)# ip flow monitor my-monitor {input|output}
```

An example of NetFlow collector is the NetFlow Analyzer by ManageEngine which you can try in your team project.

https://www.manageengine.com/products/netflow/download-free.html



Today, instead of separate logs, SIEM (Security Information and Event Management) solution is getting common to provide unified real-time correlation of all logs and threat alerting.